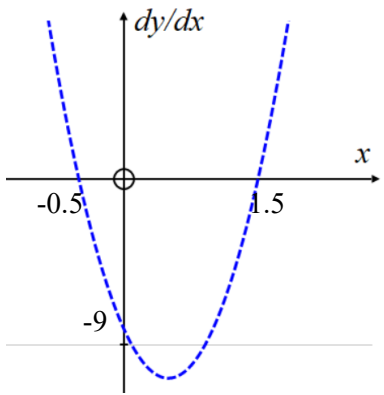
Question			Answer	Marks	AOs	Guidance	
7	(a)		$\frac{dy}{dx} = 12x^2 - 12x - 9$	M1	1.1a	**DR** Attempt to differentiate seen	
			When $\frac{dy}{dx} = 12x^2 - 12x - 9 = 0$ $3(2x+1)(2x-3) = 0$ so $x = -0.5, \quad 1.5$	M1 (dep) A1	1.1a 1.1a	Attempt to solve their $\frac{dy}{dx} = 0$ Both values seen – may be indicated on the graph	
				B1 [4]	1.1	Correct shape through $(0,-9)$	SC For cubic graph of the function drawn with M0M0A0 allow SC1 for correct shape with minimum when $x = 1.5$, and maximum when $x = -0.5$
7	(b)		Min point of gradient function when $\frac{d^2y}{dx^2} = 24x - 12 = 0$ so $x = \frac{1}{2}$	M1	3.1a	**DR** Attempt to find the vertex (including completing the square or symmetry argument)	
			Gradient is decreasing for $\left\{x : x < \frac{1}{2}\right\}$	A1 [2]	2.5	Inequality correctly formed and expressed as a set. Allow either < or ≤	